

Safe Reinforcement Learning for Battery Energy Storage Participation in the Imbalance Settlement

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Abstract—The growing uncertainty in both demand and supply sides has led to more frequent and severe imbalances in power systems, creating new opportunities for flexible Battery Energy Storage System (BESS) to profit from price variability in the Imbalance Settlement (IS) market. This paper introduces a fast and Safe Reinforcement Learning (RL) framework that unifies data-driven market learning with physics-based feasibility guarantees. Instead of relying on arbitrary approximations of market dynamics, a deep neural actor-critic captures the strategic behaviors in the IS directly from data, in a fully end-to-end manner. Safety is enforced by a differentiable projection layer that maps every action onto the convex BESS feasible set, ensuring hard constraint satisfaction during both training and deployment. By separating the learning of unknown economics from the enforcement of known BESS physics within a single trainable pipeline, the proposed safe agent achieves millisecond-scale runtime and structurally safe policies. The framework leverages off-policy RL to reuse past experiences and expert-guided trajectories, accelerating convergence and avoiding short-sighted strategies. On real Belgian IS market data, our approach delivers higher profits than both RL and model-based baselines, with faster and more stable convergence among RL methods.

Index Terms—Differentiable Optimization, Real-Time Market, Safe Layer, Safe Reinforcement Learning, Trustworthy AI.

I. INTRODUCTION

The task of maintaining the Real-Time (RT) energy balance in electricity systems is becoming increasingly complex due to growing uncertainties in both supply and demand [1]. On the supply side, the growing share of weather-dependent and intermittent Renewable Energy Sources (RES), such as photovoltaic panels and wind turbines, introduces significant variability in electricity generation. On the demand side, the electrification of key sectors (including heating & cooling, transportation, and industry) makes electrical loads more dependent on unpredictable human behavior. As a result, RT imbalances between total load and generation are becoming more frequent and severe [2], [3].

To mitigate these energy imbalances and enhance grid stability, the European market design relies on the single-price Imbalance Settlement (IS). This mechanism aims to expose the market actors, known as Balance Responsible Parties (BRP), to the RT value of energy. Practically, it monetizes any energy deviation from the BRP's planned schedule using RT prices that are adjusted quarter-hourly according to current reserve activations. BRPs are thus incentivized to leverage

their flexibility (i.e., ability to quickly and reliably adjust their energy output) to help restore the grid balance [4]. By doing so, they can potentially capitalize on extreme price fluctuations in the market. This solution reduces the overall electricity system imbalance, especially since the adoption of the Electricity Balancing Guidelines [5].

In this context, modern BESS, such as lithium-ion, are gaining significant attention. Their high round-trip efficiency, rapid ramping capabilities, and steadily declining costs make them well-suited for inter-temporal energy arbitrage in the IS [6]. This has led to growing interest in the development of advanced strategies to maximize the BESS profitability in short-term electricity markets [7]. This task remains notoriously difficult due to both the high volatility of RT prices and the very short time delay to make decisions (i.e., an action must be taken within seconds).

In [8], the joint participation of a BESS in the day-ahead market and Imbalance Settlement (IS) is developed. The same problem is solved using a stochastic model that considers price uncertainties in [9]. In [10], the day-ahead schedule of a storage system is re-optimized with a receding horizon for a US-style RT market. These works employ a Linear Program (LP)-based model of the BESS, requiring significant approximations of the inherently non-linear characteristics [11]. A high accuracy mixed-integer LP model for the BESS is used in [12]. Alternatively, a non-linear model predictive control is developed in [13]. Yet, mixed-integer and non-linear formulations, while more accurate, impose a computational burden that may not comply with RT decision-making.

In addition, all these contributions disregard the impact of the BESS decisions on the market price. This may lead to over-optimistic decisions due to high price sensitivity in the IS [14]. In order to explicitly account for price-maker effects, a solution is to incorporate the RT market clearing into the BESS optimization. This gives rise to a bi-level model, which is recast as a single-level mixed integer linear program in [15]. To account for the uncertain grid balancing conditions, the model is embedded into a single-time-horizon robust optimization framework. The methodology is extended in [16] to include degradation costs along a multi-time-step horizon, thus improving the long-term expected profits. Yet, such model-based formulations are inherently plagued with imperfect information on the reserve supply curve and the future system imbalance. This may lead to a poor parameterization of the optimization model, which may strongly affect the quality of the resulting actions [17].

To address these parameterization challenges and compu-

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This research was carried out in the context of the Energy Transition Fund, DISCRETE project, supported by the FPS Economy, SMEs, Self-Employed and Energy, Belgium

tational tractability issues, another line of research explores end-to-end machine learning methods. In particular, with the advent of deep Neural Networks (NN), decision policies can be learned directly from available raw inputs, bypassing the need for analytical formulations that may struggle with scalability and expressiveness. This is particularly relevant in our setting, where the IS clearing mechanism and the private strategies of market participants remain largely unobservable and unpredictable. A possible solution is provided by NN-constrained optimization [18], where market dynamics are embedded as NN-based constraints within the BESS optimization model. However, this approach remains impractical for real-time operation, as the implicit NN constraints lead to a non-convex problem that is computationally demanding to solve. Alternatively, Reinforcement Learning (RL) has been investigated in [19], by training a soft actor-critic algorithm to perform the temporal arbitrage of a BESS in the IS. Once trained on historical data, such RL agents take fast decisions, but with no guarantee on the solution quality and its physical feasibility (i.e., there is no guarantee that the prescribed solutions do not violate physical constraints). To address this challenge, this paper builds upon a new paradigm, known as safe RL [20], [21], which investigates how we can guide the NN-based agent's training to improve the reward (or objective) while complying with hard constraints.

One research direction in safe RL explores learning with soft constraints, e.g., by subtracting a fixed penalty from the reward whenever a violation occurs during training [22]. The idea is extended by the Lagrangian multiplier method which dynamically adjusts the penalty based on the severity of violations to trade off feasibility and reward [23]. Other approaches iteratively reduce violations using differentiable gradient descent [24], or apply barrier functions that penalize actions as they approach constraint boundaries [25].

To improve upon penalty-based or auxiliary-loss methods that bias rewards and can fail to enforce feasibility at deployment, an alternative line of research focuses on projections, which map unsafe actions onto the feasible region [26]–[28]. Recent works in the energy domain adopt this idea, showing that differentiable projection layers can enforce feasibility during both training and deployment [29], [30], or act as post-training corrections under simplified “policy-friendly” constraints [31]. While effective at enforcing feasibility, these approaches incur computational overhead, making them less suited for real-time market applications.

Our work advances this line of research by introducing a physics-informed convex projection that enforces known BESS constraints, while leaving the learning of market dynamics entirely to a NN. Since the private decisions and information of other market participants are not observable, the proposed framework aims to infer useful representations of the market dynamics, such as action–price feedback and strategic interactions, directly from aggregated price and cleared volume data. To this end, we adopt an actor–critic architecture, which is particularly well suited for continuous control problems as it jointly learns a policy (actor) and a value function (critic) to optimize decision-making under uncertainty. The actor NN is augmented with a differentiable projection layer

that ensures that all BESS operational constraints are satisfied. This explicit decoupling between (i) learning the unknown economics and (ii) enforcing the known physics, within a single trainable pipeline, preserves convexity of the safety step, thus enabling millisecond-scale runtime and overcoming a key limitation of existing approaches.

The main contributions of this paper are summarized below.

- 1) We develop a fast and safe deep RL framework for the sequential decision-making process of a price-maker BESS participating in the IS market. The proposed agent learns to act optimally under partial observability by exploiting the statistical regularities embedded in historical price and volume data, eliminating the need for approximate analytical market models. Safety is enforced by augmenting the agent with a differentiable optimization layer that projects every action onto the BESS feasible set. This makes the architecture fully end-to-end trainable while guaranteeing zero infeasible actions in both training and deployment.
- 2) We incorporate expert trajectories and optimization-based high-reward actions into the training process. These are used to bootstrap the policy, which is then refined online by the safe RL agent using self-collected data, enabling the agent to adaptively adjust its risk profile in response to time-varying market conditions.
- 3) To address the inherent partial observability of the IS market, particularly regarding the hidden strategies of other market participants, we enhance the agent architecture with an advanced Recurrent NN (RNN). Specifically, we design a memory-augmented model, based on Long Short-Term Memory (LSTM) neurons, that reconstructs a latent representation of the hidden state by jointly encoding both historical observations and future forecasts.

We demonstrate our approach in a case study using Belgian market data, showing that the trained safe RL agent makes decisions within 1 to 2 milliseconds, with 0% infeasible actions, and significantly higher ex-post profits compared to penalty-based and unconstrained RL baselines. While validated in IS, the proposed agent is broadly transferable to other physical assets and electricity market settings.

The structure of the paper is as follows. We start in Section II by defining the real-time BESS control problem in the IS market, focusing on the objective of profit maximization under uncertainties. Then, we focus in Section III on the solution method, introducing the RL agent augmented with a safety layer, along with the training procedure. Afterwards, we briefly present in Section IV the different benchmark strategies, including other safe RL methods, and both model-free and model-based solutions. The case study in Section V compares the performance (e.g., training time and ex-post profits) on real data from the Belgian market. Section VI concludes by drawing the main observations and perspectives.

II. PROBLEM FORMULATION

In this paper, we consider a BESS, acting as a BRP, participating in the single-price IS. In this market segment,

BRPs are financially settled based on a uniform imbalance price λ^{SI} for each deviation between their scheduled and their actual energy consumption or production during a 15-minute period. This real-time price λ^{SI} reflects the cost incurred by the system operator to correct the country-wide System Imbalance (SI), which arises from the net aggregated deviation of all BRPs. In practice, λ^{SI} captures the marginal cost of activating reserves in the balancing market. The SI is thus highly unpredictable, as it is affected by the erratic behavior of all consumers, the weather-dependent renewable generation and the strategic behavior of market agents.

If the imbalance of an individual BRP helps reducing the SI, it is rewarded at the imbalance price λ^{SI} . Conversely, if it worsens the SI, it is penalized at the same price. In practice, actively participating in the IS is complex due to the fast fluctuations of the grid's balancing needs, translating in volatile RT prices.

We represent in (1) the risk-neutral formulation of the problem using a stochastic bi-level optimization model. The upper-level objective (1a) maximizes the expected net profit over the look-ahead horizon $\mathcal{T} = [t_1, t_1 + H]$, based on the predicted distribution $\mathbb{P}_{SI}$ of future system imbalances. Here, t_1 denotes the start of the upcoming decision interval, with a resolution of $\Delta t = 15$ minutes. The profit per time step is defined as the difference between the market revenues and the non-linear BESS degradation costs Γ_t . Market revenues are obtained by multiplying the BESS energy output $e_t = p_t \cdot \Delta t$ with the imbalance price λ_t^{SI} . Given the small size of the IS market (typically on the order of a few tens of MW), the BESS's actions can affect the price λ_t^{SI} . As a result, the clearing mechanism of the balancing market must be explicitly modeled for each $t \in \mathcal{T}$, which is addressed in the lower-level optimization problems (1b).

$$u^* \in \arg \max_{u \in \mathcal{U}(I^*)} \mathbb{E}_{\mathbf{si} \sim \mathbb{P}_{SI}} \left[\sum_{t=t_1}^{t_1+H} (\lambda_t^{SI}(l_t^*) \cdot e_t - \Gamma_t) \right] \quad (1a)$$

$$\text{s.t. } l_t^* \in \arg \min_{l_t \in \mathcal{L}(si_t)} \mathbb{E}_{si_t \sim \mathbb{P}_{SI_t}} \left[\sum_{i \in \mathcal{I}_t} \Lambda_{i,t}^+ \cdot s_{i,t}^+(si_t, e_t) - \sum_{j \in \mathcal{J}_t} \Lambda_{j,t}^- \cdot s_{j,t}^-(si_t, e_t) \right], \quad \forall t \in \mathcal{T} \quad (1b)$$

where $\mathbb{P}_{SI}$ is the predicted joint distribution of system imbalances $\mathbf{si} = \{si_t\}_{t \in \mathcal{T}}$ over the horizon. The uncertainty in each si_t propagates through the lower-level market-clearing problem in (1b) to define the imbalance prices $\lambda_t^{SI}(l_t^*) := \lambda_t^{SI}(l_t^*(si_t))$.

The upper-level feasible set $\mathcal{U}(I^*)$ in (1a) encodes the BESS techno-economic constraints, while restricting actions based on the expected outcomes $I^* = \{l_t^*\}_{t \in \mathcal{T}}$ of the balancing market over the look-ahead horizon.

The lower-level problems (1b) minimize the expected cost of balancing the grid at each time step $t \in \mathcal{T}$, under the (predicted) marginal distribution $\mathbb{P}_{SI_t}$ of the SI (in MW) at time t . These costs result from the activation of balancing reserves by the system operator, and are aggregated into a single marginal activation price λ_t^{SI} .

The lower-level feasible set $\mathcal{L}(si_t)$ depends on the balancing offers of all Balancing Service Providers (BSP). The decision variables l_t are the cleared quantities for upward $s_{i,t}^+$ and downward $s_{j,t}^-$ regulations, which depend on the predicted $si_t \sim \mathbb{P}_{SI_t}$ and the BESS energy deviation e_t . Here, $i \in \mathcal{I}_t$ and $j \in \mathcal{J}_t$ correspond to the sets of individual balancing offers in each direction. These decisions are influenced by the bidding prices $\Lambda_{i,t}^+$ and $\Lambda_{j,t}^-$, which represent the costs for activating the reserves. The resulting merit-order curve is represented in Fig. 1.

In practice, these parameters $\Lambda_{i,t}^+$ and $\Lambda_{j,t}^-$ are only partially observable by individual BRPs, which can lead to inaccuracies and suboptimal decisions. Furthermore, the formulation assumes that the system operator perfectly compensates for SI, which may not hold in real-world conditions and can degrade the quality of the upper-level decisions u^* . Finally, the resulting non-linear non-convex bi-level problem (1) is computationally demanding and difficult to solve, particularly given the limited time to take decisions in the IS market. Since decisions need to be made every 15 minutes in accordance with the IS resolution, a fast inference speed is crucial, as any additional computation time translates into idle operation and lost revenues, a limitation of optimization-based approaches.

To address these limitations, we propose a solution based on a safe RL framework. The RL algorithm adapts to uncertain, non-convex dynamics, i.e., capturing the IS market clearing process, the strategic behavior of competing agents, and the feedback between its own actions and market prices. We further provide guarantees on the solution feasibility by encoding hard constraints of the known BESS physical dynamics within a computationally efficient optimization layer.

Our fast and reliable two-stage RL agent is represented in Fig. 2. This agent is fully differentiable, forming a consistent end-to-end trainable system. During deployment, at each time step t , a deep NN is fed with relevant contextual information s_t to yield the BESS output power $\hat{p}_t$, which is then corrected into p_t^* by the safe layer.

III. METHODOLOGY

To enable the use of RL, we recast the problem as a Markov Decision Process (MDP) in Subsection III-A. Building on this formalism, we introduce our RL agent in Subsection III-B. Then, in Subsection III-C, we enhance the agent with a safety

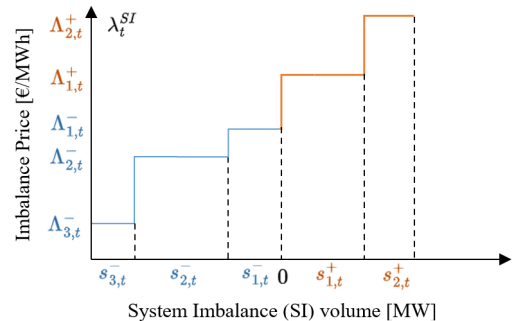


Fig. 1. Single imbalance prices per volume levels for one quarter-of-an-hour, made publicly available by the system operator.

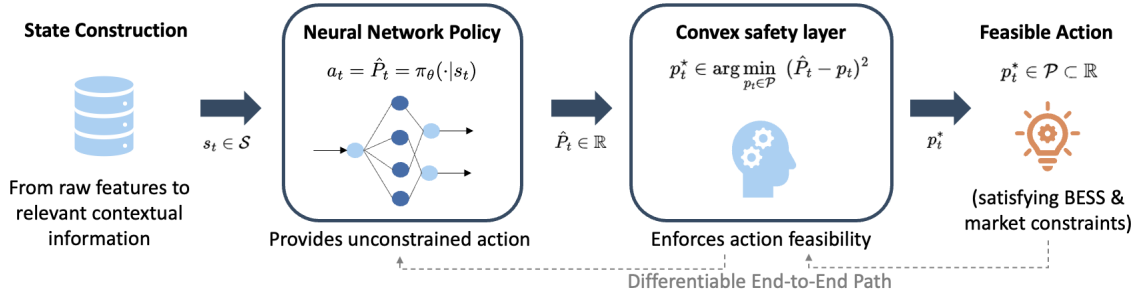


Fig. 2. Architecture of the proposed safe RL framework. A NN policy outputs raw actions, which are projected onto a feasible action set via a convex projection layer that enforces BESS and market constraints. It is fully differentiable and cohesively forms an end-to-end trainable system, allowing stable training while ensuring safe actions at deployment.

layer to ensure the feasibility of BESS actions. Finally, in Subsection III-D, we discuss the training procedure of the resulting safe RL agent.

A. Formulation as a Markov Decision Process

To enable the application of the RL paradigm, the energy arbitrage problem is formulated as a Markov Decision Process (MDP) that can be mathematically described as a 6-tuple $\langle \mathcal{S}, \mathcal{A}, \mathbb{P}_{\mathcal{E}}, r, \rho_0, \gamma \rangle$. In this setting, ρ_0 represents the distribution of possible conditions at the start of a training episode. Then, at each time step t , an agent receives a partial observation of the environment's state $s_t \in \mathcal{S}$ and selects an action $a_t = \hat{P}_t \in \mathcal{A}$ by following a (deterministic or stochastic) policy $\pi(a_t | s_t)$. The environment then evolves to a new state s_{t+1} and the agent receives a reward r_{t+1} that evaluates the quality of the action, according to the transition dynamics $(s_{t+1}, r_t) \sim \mathbb{P}_{\mathcal{E}}(\cdot | s_t, a_t)$.

The policy $\pi(a_t | s_t)$ is optimized to maximize the agent's expected return over the horizon of T time steps. This objective is captured by the action-value function $Q^\pi(s_t, a_t) = \mathbb{E}_{\tau \sim \mathbb{P}_\tau} \left[\sum_{k=t}^T \gamma^{(k-t)} r_k | s_t, a_t \right]$, where $\gamma \in [0, 1]$ is the discount factor. The expectation $\mathbb{E}_{\tau \sim \mathbb{P}_\tau}$ is taken over the trajectories $\tau = (s_t, a_t, \dots, s_{T+1}) \sim \mathbb{P}_\tau$ generated by taking action a_t in state s_t and following the policy π thereafter. It should be noted that the policy is thus trained to generate actions with a long-term perspective, rather than adopting a short-sighted greedy strategy that prioritizes immediate rewards.

1) *State and Observation Set*: The state of the environment represents all the relevant information needed for the BESS to make optimal decisions. The state set $\mathcal{S}$ is composed of both the BESS and the IS market. Since the BESS ramp rates are very fast relative to the IS time scale [32], its operating state can be fully characterized by its State-of-Charge (SoC).

Regarding the IS market, the BESS only has access to a partial and uncertain observation of the relevant variables, which omits the financial strategy of other BRPs and the true price function $\lambda_t^{SI} = \delta_t(\text{SI})$, i.e., the true model (1b), used by the system operator to activate balancing reserves. To address this partial observability in the IS market, two complementary strategies are implemented.

First, to deal with the uncertainty in both future imbalance volumes (which is critical for ensuring that BESS actions restore the system's balance) and in imbalance prices (which

determine whether price spreads are sufficient for generating profits), we train a probabilistic forecaster. The model, built upon a deep transformer from [14], is designed to provide a distributional view of possible future SIs. These volumes are then converted into prices using the estimated $\lambda_t^{SI} = \hat{\delta}_t(\text{SI})$.

Second, as explained in Section III-B2, we use a RL agent equipped with memory. This allows the agent to keep track of past observations through an internal hidden state, helping it to infer the relevant unobserved dynamics of the environment. The state space $\mathcal{S}$ is defined by the following signals.

- The (normalized) state of charge of the BESS before t_1 : $\text{soc}_{t_0} \in [0, 1]$.
- The predicted distribution $\mathbb{P}_{\mathcal{S} \mathcal{I}_{t_1}}$ of the SI for the upcoming time step t_1 under the form of $|\mathcal{Q}| = 11$ quantiles, i.e., $\{\lambda_{t_1}^{SI, q}\}_{q \in \mathcal{Q}}$ with $\mathcal{Q} = \{0.01, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95, 0.99\}$.
- The projected IS prices for the upcoming time step t_1 for different values of the SI (from -600 MW to 600 MW with a 100 MW granularity) based on the (incomplete) information given by the system operator.
- The predicted median value of the SI volumes and prices for the next 10 time steps $[t_1, t_{10}]$.
- The predicted standard deviation of the SI volumes and prices for the next 10 time steps $[t_1, t_{10}]$.
- 4 calendar features encoding the quarter-hour and hour of the day using two pairs of phase-shifted sine functions, allowing to capture recurring daily patterns.

2) *Action Set*: The action set $\mathcal{A}$ is defined by the active (dis)charging power of the BESS for the upcoming 15-minute time step t_1 . The action space, from maximum BESS charge to maximum discharge power, is either continuous $\mathcal{A} = [-1, 1]$ or discrete $\mathcal{A} = \{a_i\}_{i=1}^n$, depending on the RL algorithm.

3) *Reward Function*: At each 15-minute interval Δt , the immediate reward r_t represents the net profit (or loss) obtained by the BESS, jointly accounting for market revenues (or costs) and degradation effects. The cumulative return over an episode, defined as the sum of interval-level rewards, inherently captures inter-temporal arbitrage opportunities, since charging actions at earlier intervals can create value by enabling profitable discharging in subsequent intervals.

Formally, the immediate reward is defined in (2) as the Imbalance Settlement revenue (or cost) associated with the

exchanged energy e_t , minus the degradation cost Γ_t :

$$r_t = e_t \cdot \delta_t - \Gamma_t(R_c, e_t). \quad (2)$$

The exchanged energy e_t [MWh] depends on the BESS action $\hat{P}_t$ (charging or discharging power) and is adjusted for charge η^{ch} and discharge η^{dis} efficiencies, as detailed in (3):

$$e_t = \begin{cases} \frac{\hat{P}_t}{\eta^{\text{ch}}} \cdot \Delta t, & \text{if } \hat{P}_t < 0 \quad (\text{charging}) \\ \eta^{\text{dis}} \cdot \hat{P}_t \cdot \Delta t, & \text{if } \hat{P}_t > 0 \quad (\text{discharging}). \end{cases} \quad (3)$$

Finally, the degradation cost Γ_t is expressed in (4):

$$\Gamma_t(R_c, e_t) = R_c \cdot \Phi(\beta_t) \quad (4)$$

where R_c denotes the battery replacement cost, and Φ is the cycle depth stress function specific to lithium-ion batteries [33]. This function Φ is the percentage of life loss caused by a cycle of depth β . As proposed in [34], β is approximated by the energy output within a control time interval, i.e., $\beta_t = e_t$, with $\Phi(\beta_t)$ given by:

$$\Phi(\beta_t) = 5.24e^{-4} e_t^{2.03}. \quad (5)$$

4) *Environment*: The environment simulates the interaction between the price-maker BESS and the IS market by returning new states s_{t+1} and rewards r_t in response to the BESS's actions. It captures both the techno-economic dynamics of the BESS and the market clearing mechanism of the IS, which determines real-time prices λ_t^{SI} .

Beyond the degradation costs, the physical constraints of the BESS include the SoC evolution. The latter is only determined by the action p_t selected by the RL agent and is computed according to (6), where e_t is derived from (3).

$$soc_{t+1} = soc_t - e_t \quad (6)$$

If soc_{t+1} exceeds the BESS physical bounds, a clipping operation is achieved to ensure that $soc_{t+1} \in [0, 1]$.

Inspired by [35], we refine conventional BESS models that approximate feasible power limits with fixed box constraints. In practice, electrochemical characteristics restrict charging and discharging: the maximum achievable power decreases at both low and high SoC levels due to internal resistance and voltage limitations. To capture this effect, the feasible region is modeled with parabolic charge and discharge envelopes, which preserve convexity and therefore tractability. Specifically, the upper and lower bounds on charging/discharging power are expressed as quadratic functions of the next-period SoC:

$$\hat{P}_t \leq -1.4732 soc_{t+1}^2 + 1.3518 soc_{t+1} + 0.6393 \quad (7)$$

$$\hat{P}_t \geq 1.4732 soc_{t+1}^2 - 1.3518 soc_{t+1} - 0.6393 \quad (8)$$

where the coefficients are obtained by least-squares regression on the detailed electrochemical model reported in [35]. These quadratic bounds define the feasible charge/discharge envelope used in our projection layer, ensuring both accuracy and convexity compared to static box constraints.

The IS is modeled by imbalance prices λ_t^{SI} , from which the BESS profit is computed. These prices are determined by the intersection of the supply curve (based on the actual bidding prices $\Lambda_{i,t}^+$ and $\Lambda_{j,t}^-$ submitted by all BSPs in

the balancing market) and the activated reserves $s_{i,t}^+$ and $s_{i,t}^-$, which are used to offset the observed SI, accounting for BESS actions. Indeed, BESS actions p_t affect the SI, which, in turn affects λ_t^{SI} and thus the BESS rewards.

B. RL agent

The RL problem consists in determining the optimal policy π^* that maximizes the expected return $J(\pi)$ from the initial state distribution ρ_0 , i.e., $\pi^* \in \arg \max_{\pi} J(\pi)$ with

$$J(\pi) = \mathbb{E}_{s_0 \sim \rho_0, a_0 \sim \pi(\cdot | s_0)} [Q^{\pi}(s_0, a_0)] \quad (9)$$

where ρ_0 defines the probability for drawing the initial state s_0 , representing here the initial SoC of the battery along with the grid and market conditions at $t = t_0$.

In this paper, a parameterized policy π_{θ} is learned using the expressive capacity of deep NNs (with weights θ). However, learning such a policy in the context of the IS market is challenging because buying actions typically lead to negative immediate rewards, which can bias the policy toward short-term profit-maximizing (i.e., selling) actions, thereby impeding the discovery of optimal long-term strategies via inter-temporal arbitrage. To address this issue, a high-quality exploration mechanism is essential to encourage the agent to explore actions that may appear suboptimal in the short term but lead to higher cumulative rewards over time.

Consequently, the Soft Actor-Critic (SAC) agent is particularly well-suited, as its entropy-regularized objective promotes diverse and persistent exploration, enabling the agent to uncover and exploit profitable arbitrage opportunities despite misleading immediate feedback.

1) *Soft Actor-Critic*: The objective of model-free, off-policy SAC is to learn a stochastic policy $\pi_{\theta}(a|s)$ that maximizes the expected cumulative rewards plus an entropy term $\mathcal{H}(\pi_{\theta})$, which encourages exploration and prevents premature convergence to a suboptimal deterministic policy [36]. The entropy of the policy at state s_t is given by:

$$\mathcal{H}(\pi_{\theta}(\cdot | s_t)) = \mathbb{E}_{a_t \sim \pi_{\theta}(\cdot | s_t)} [-\log \pi_{\theta}(\cdot | s_t)]. \quad (10)$$

The entropy-regularized objective of the SAC agent is

$$J(\pi_{\theta}) = \mathbb{E}_{\tau \sim \mathbb{P}(\tau | \pi_{\theta})} \left[R(\tau) + \alpha \sum_{t=0}^T \mathcal{H}(\pi_{\theta}(\cdot | s_t)) \right], \quad (11)$$

where α is a temperature parameter that controls the trade-off between return maximization and policy entropy.

In practice, the SAC objective (11) is approximated by a surrogate that is more computationally efficient and compatible with off-policy training.

First, to bypass the need to compute expectations over full trajectories τ , SAC introduces learned critics $Q_{\phi_1}(s, a)$ and $Q_{\phi_2}(s, a)$, parameterized by separate NNs with weights ϕ_1 and ϕ_2 . Second, to improve the learning stability, SAC uses the clipped double- Q trick, where the minimum of the two Q -value estimates is taken to reduce overestimation bias,

i.e., $Q_\phi(s, a) = \min(Q_{\phi_1}(s, a), Q_{\phi_2}(s, a))$. The entropy-augmented Q -functions are defined as:

$$Q_{\phi_i}(s, a) = \mathbb{E}_{\tau \sim \mathbb{P}(\tau|\pi_\theta)} \left[\sum_{k=0}^{T-t} \gamma^k r_{t+k} + \alpha \sum_{k=1}^{T-t} \gamma^k \mathcal{H}(\pi(\cdot|s_{t+k})) \mid s_t = s, a_t = a \right], i \in \{1, 2\}. \quad (12)$$

The final surrogate objective used for policy updates is then:

$$\hat{J}(\pi_\theta) = \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta(\cdot|s)} [\alpha \log \pi_\theta(a|s) - Q_\phi(s, a)], \quad (13)$$

which is computed using mini-batches sampled from a replay buffer $\mathcal{D}$ that stores off-policy transitions (s_t, a_t, r_t, s_{t+1}) collected during interactions with the environment.

In this paper, the stochastic policy $\pi_\theta(a|s)$, also called the actor, is implemented as a NN parameterized by θ . It yields a Gaussian distribution over actions $a_t \sim \mathcal{N}(\mu_\theta(s_t), \sigma_\theta^2(s_t))$, characterized by a state-dependent mean $\mu_\theta(s_t)$ and standard deviation $\sigma_\theta(s_t)$.

In practice, $\pi_\theta(a_t|s_t)$ needs to be sampled to generate a single action $a_t = \hat{P}_t$. A random sampling $\hat{P}_t \sim \mathcal{N}(\cdot)$ is carried out during training to promote exploration, while the mean $\hat{P}_t = \mu_\theta(s_t)$ is used in validation and test sets to ensure a consistent BESS behavior.

However, the sampling operation is inherently non-differentiable. To allow efficient gradient-based optimization (see Section III-D), SAC uses the reparameterization trick, expressing the action as a differentiable function of a noise variable ϵ , i.e., $\hat{P}_t = f_\theta(\epsilon, s_t)$. This allows to move the sampling operation on a variable ϵ that is outside the computational graph, making the NN agent differentiable with respect to the policy parameters θ . Here, we leverage the fact that the policy output π_θ is a Gaussian distribution $\mathcal{N}(\mu_\theta, \sigma_\theta)$ by using:

$$\hat{P}_t = \tanh(\mu_\theta(s_t) + \sigma_\theta(s_t) \cdot \epsilon) \in [-1, 1], \quad (14)$$

with $\epsilon \sim \mathcal{N}(0, 1)$.

2) *Recurrent Agent for Partial Observability*: Due to limited observability in the IS market, i.e., from the uncertain future SIs and the hidden true price formation mechanism $\lambda_t^{SI} = \delta_t(SI)$, a memory-based NN model is used to model the policy π_θ and the critic. This turns the agent into a Partially Observable MDP (POMDP) solver, enabling to uncover and keep in memory the relevant underlying market dynamics.

Practically, we use an advanced sequence-to-sequence (Seq2Seq) architecture based on Long Short-Term Memory (LSTM) NNs. LSTM is a type of recurrent NN capable of maintaining an internal hidden state that captures and memorizes temporal dependencies. This internal memory enables the agent to infer market dynamics that are not directly observed. The resulting architecture is shown in Fig. 3.

In our setup, an LSTM-based encoder processes a sequence of LB look-back observations. At each $t \in [t_{-LB} : t_0]$, it updates its hidden state h_t^{enc} (short-term memory) and internal cell state c_t^{enc} (long-term memory), which together carry the relevant temporal context (15).

$$h_t^{\text{enc}}, c_t^{\text{enc}} = \text{LSTM}^{\text{enc}}(x_t^{\text{h}}, h_{t-1}^{\text{enc}}, c_{t-1}^{\text{enc}}), \quad \forall t \in [t_{-LB}, t_0] \quad (15)$$

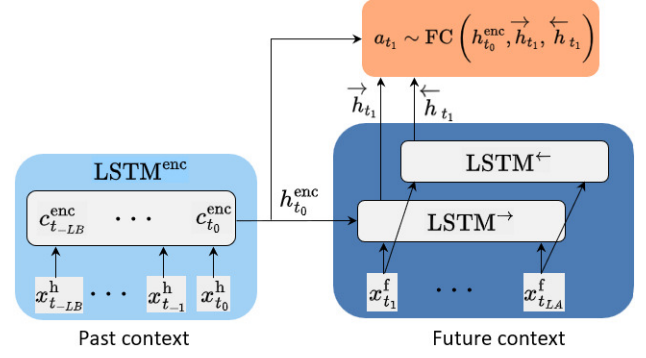


Fig. 3. Sequence-to-sequence LSTM-based model used to infer unobserved dynamics of the partially observable environment.

where x_t^{h} is the input feature vector at time t (Section III-A1), and t_0 is the current time step.

To capture the full structure of future context (e.g., predicted future SI conditions), we employ a bidirectional LSTM decoder (16) that processes the sequence of LA look-ahead forecasts $x_{t'}^{\text{f}}$. To ensure that the decoder leverages the temporal summary learned by the encoder, we initialize the forward LSTM component $\text{LSTM}^{\rightarrow}$ with the encoder's final hidden and cell states, i.e., $\vec{h}_{t_1} = h_{t_0}^{\text{enc}}$ and $\vec{c}_{t_1} = c_{t_0}^{\text{enc}}$.

$$\begin{aligned} \vec{h}_{t'}, \vec{c}_{t'} &= \text{LSTM}^{\rightarrow}(x_{t'}^{\text{f}}, \vec{h}_{t'-1}, \vec{c}_{t'-1}) \\ \overleftarrow{h}_{t'}, \overleftarrow{c}_{t'} &= \text{LSTM}^{\leftarrow}(x_{t'}^{\text{f}}, \overleftarrow{h}_{t'+1}, \overleftarrow{c}_{t'+1}) \end{aligned} \quad \forall t' \in [t_1, t_{LA}] \quad (16)$$

We extract the forward and backward states of decision time t_1 and concatenate them into a single context vector:

$$h_{t_1}^{\text{dec}} = [\vec{h}_{t_1}; \overleftarrow{h}_{t_1}] \quad (17)$$

Finally, the model output is derived from the fused encoder and decoder representations via a fully connected (FC) layer. For the actor, it yields the action

$$a_{t_1} \sim \text{FC}(h_{t_0}^{\text{enc}}, h_{t_1}^{\text{dec}}). \quad (18)$$

The resulting agent is conditioned on both historical and anticipated future information, thereby relaxing the Markov assumption and better handling partial observability.

C. Differentiable Projection of the Action

The BESS output power $\hat{P}_t$ provided by the NN actor may not be feasible, i.e., $\hat{P}_t$ may not comply with the constraints of the problem. As a result, BRPs may face unpredictable financial losses by failing to follow their optimized policy. To tackle this problem, the NN output $\pi_\theta(a_t|s_t)$ is here combined with a correction layer $\mathcal{C} : \mathbb{R} \rightarrow \mathbb{R}$ to improve the quality of the policy, leading to a two-stage global model

$$\mathcal{M}_\theta(s_t) = \mathcal{C} \circ (\hat{P}_t \sim \pi_\theta(\cdot|s_t)) = p_t^* \quad (19)$$

where the correction layer $\mathcal{C}$ consists in a differentiable projection of predicted actions $\hat{P}_t$ onto the feasible set $\mathcal{P} \subset \mathbb{R}$, leading to the optimal decision $p_t^* = \mathcal{C}(\hat{P}_t) \in \mathcal{P}$.

Unlike penalty- or barrier-based safe RL, the correction $\mathcal{C}$ leaves the reward function untouched. Safety is instead ensured

by this differentiable projection layer $\mathcal{C}$ that maps every raw action onto the BESS feasible set. This safe layer enforces physical constraints in a transparent, interpretable way while still allowing gradients to flow through the correction step. As a result, actor and critic updates are confined to a convex manifold defined by BESS physics, which stabilizes learning, accelerates convergence, and yields higher-quality policies.

However, the safe layer is often computationally inefficient, as it requires solving an optimization model with the same complex constraints as the original problem [37]. Here, we bypass this issue by observing that action feasibility depends only on a subset of constraints. Only the BESS dynamics affects the feasibility of BESS actions. Therefore, the IS market dynamics do not need to be included in the projection step. The BESS constraints (6)–(8) define a convex feasible set, so the projection reduces to a small quadratic program solvable by standard methods [38]. By differentiating through its KKT conditions, we preserve end-to-end gradient flow while ensuring feasibility during both training and deployment.

Overall, we formulate the projection problem of the predicted action $\hat{P}_t$ at time step t into the feasible domain as:

$$p_t^* \in \arg \min_{p_t} (\hat{P}_t - p_t)^2 \quad (20a)$$

$$\text{s.t. } 0 \leq \text{soc}_{t+1} \leq 1 \quad (20b)$$

$$\text{soc}_{t+1} = \text{SOC}_t - p_t \cdot \Delta t \quad (20c)$$

$$-1 \leq p_t \leq 1 \quad (20d)$$

$$(7), (8) \quad (20e)$$

where SOC_t is the BESS energy content before any deviation in the IS, while Δt is the time duration of the IS market segment (i.e., 15 minutes).

The projection layer guarantees 1-step feasibility by enforcing SoC bounds and power limits based on the current state. Longer-term optimality is not directly encoded in these constraints but is learned through the RL paradigm, since the expected return is defined over the full horizon. In practice, this combination prevents dead-end trajectories and ensures that all decisions remain feasible during both training and deployment, as confirmed in the case study (Section V).

D. Training of the safe RL agent

The safe SAC architecture is illustrated in Fig. 4. In the following, we describe the training procedure used to optimize the parameters θ and ϕ of the policy and Q -function NNs.

1) *Off-Policy Gradient Updates*: The goal of the safe RL agent is to maximize the expected return $J(\mathcal{M}_\theta(s_t))$ defined in (13) where the traditional policy $\pi_\theta(s_t)$ is replaced by our proposed safe policy $\mathcal{M}_\theta(s_t)$. This is achieved by optimizing the parameters θ of the safe policy using gradient descent, i.e., finding $\nabla_\theta J(\mathcal{M}_\theta(s_t))$. By applying the chain rule, we have:

$$\nabla_\theta J(\mathcal{M}_\theta(s_t)) = \frac{\partial J(\mathcal{M}_\theta(s_t))}{\partial p^*} \frac{\partial p^*}{\partial \hat{P}} \frac{\partial \hat{P}}{\partial \theta}. \quad (21)$$

Solving (21) requires to compute the derivative of the optimal solution p^* of the safe layer with respect to the input $\hat{P}$, i.e., $\partial p^* / \partial \hat{P}$, where p^* depends on the argmin operation. This task is non-trivial, but we can leverage the fact that

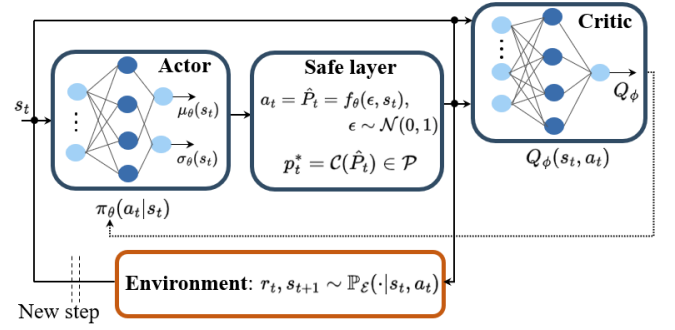


Fig. 4. Safe Soft Actor-Critic framework.

the projection problem is convex, as it features a quadratic objective function subject to convex constraints. This convex structure allows to use recent advancements in differentiable convex optimization [39].

Practically, the differentiable projection layer is implemented using CVXPYLayers [40], which enables end-to-end gradients through convex optimization problems. The convex program defining the BESS feasibility set is first reformulated as a conic program [41]. Gradients are then obtained via implicit differentiation of the KKT conditions, without backpropagating through solver iterations. This provides exact and numerically stable derivatives of the projection step, while keeping the additional computational cost negligible compared to the forward pass. This allows to automatically compute the gradients of the solution $\partial p^* / \partial \hat{P}$ analytically.

Overall, the policy gradient (21) can thus be expanded as in (22), by using the approximated equation of the entropy-regularized return from (13):

$$\nabla_\theta J(\mathcal{M}_\theta(s_t)) = \mathbb{E}_{s_t \sim \mathcal{D}, \epsilon \sim \mathcal{N}(0,1)} \left[\nabla_\theta (\alpha \log \pi_\theta(p_t^* | s_t) - Q_\phi(s_t, p_t^*)) \right] \quad (22)$$

where p_t^* is treated as a differentiable function of θ via the reparameterized action $\hat{P}_t = f_\theta(\epsilon, s_t)$ and the projection operator $\mathcal{C}$, i.e., $p_t^* = \mathcal{C}(\hat{P}_t) = f_\theta(\epsilon, s_t)$. The gradient can thus be expanded using the chain rule as:

$$\nabla_\theta J(\mathcal{M}_\theta(s_t)) = \mathbb{E}_{s_t, \epsilon} \left[\nabla_{p_t^*} (\alpha \log \pi_\theta(p_t^* | s_t) - Q_\phi(s_t, p_t^*)) \cdot \nabla_{\hat{P}_t} \mathcal{C}(\hat{P}_t) \cdot \nabla_\theta f_\theta(\epsilon, s_t) \right]. \quad (23)$$

The critic network $Q_\phi(s_t, p_t^*)$ is trained using the Prioritized Experience Replay (PER) technique, which increases the replay frequency of transitions expected to yield higher learning progress. This progress is measured by the critic's loss, known as the soft Bellman residual [36].

2) *Expert-guided Exploration*: To further integrate expert knowledge into the learning process, we introduce a guided exploration strategy during the early stages of training. Specifically, an offline dataset of expert-informed trajectories is constructed, combining (i) heuristic rule-based actions derived from imbalance price profiles (e.g., where the BESS charges during the lowest 25% and discharges during the highest 25% of forecasted prices) and (ii) trajectories generated by a simplified MPC optimization neglecting degradation and

considering a two-step look-ahead horizon. These two sources are equally weighted and jointly represent 62.5% of the initial replay buffer $\mathcal{D}$. The remaining buffer entries are filled with randomly sampled transitions to encourage diversity. Without this expert initialization, the agent often gets trapped in a trivial discharge-only policy and fails to explore profitable long-term strategies. This expert-informed warm start guides the agent toward meaningful arbitrage logic, prevents uninformative idle strategies, and enhances both learning stability and convergence speed.

IV. ALTERNATIVE APPROACHES

Our approach, called *SafeSAC*, bridges the gap between model-based methods that guarantee safety but rely on simplified market models, and model-free RL approaches that capture market dynamics but lack feasibility guarantees. This is achieved by combining data-driven learning with a differentiable projection layer for strict BESS safety. offers the advantages of both model-free RL and parametric optimization models, to obtain a fast, reliable, interpretable and data-consistent decision tool. The method is thus compared with a variety of benchmarks.

First, we test traditional model-free RL algorithms, which do not offer any guarantee on the feasibility of BESS actions.

- *SAC*, the soft actor-critic agent without the safe optimization layer.
- *DQL*, the Deep Q -learning is a RL algorithm that approximates the optimal function $Q^*(s_t, a_t)$ using a deep NN. Instead of explicitly learning a policy function, it derives the optimal policy implicitly by selecting actions that maximize the estimated Q -values for each state. As a result, no safe layer can be added to this agent.
- *TD3*, the Twin-Delayed Deep Deterministic Policy Gradient algorithm is an actor-critic algorithm designed for continuous control, which learns a deterministic policy $a_t = \pi_{\theta}^{\text{TD3}}(s_t)$ while employing an exogenous exploration strategy through action noise.

Second, we build *SafeTD3*, i.e., the TD3 agent augmented with our proposed (differentiable) projection-based layer.

Third, we examine two methods that incorporate safety into the SAC agent through alternative mechanisms, distinct from our safe layer.

- *SACrew*, a naive method to avoid infeasible schedules for the SAC agent is to enrich the reward function (2) with penalties in case of constraints violation.
- *SACpen*, a related method to improving constraint satisfaction incorporates the constraints directly into the objective function as penalty terms. In this setting, the algorithm minimizes a composite loss function ℓ_{soft} that combines the original objective with two penalty terms, which quantify the degree of violation for both equality and inequality constraints. This encourages the model to find solutions that are not only optimal with respect to the objective but also approximately feasible.

$$\ell_{\text{soft}}(a_t) = J(\pi_{\theta}) + \lambda_g \|\text{ReLU}(g(a_t))\|_2^2 + \lambda_h \|h(a_t)\|_2^2 \quad (24)$$

where $g(a_t) \leq 0$ and $h(a_t) = 0$ are the sets of inequality and equality constraints, $\lambda_g, \lambda_h > 0$ are penalty coefficients, and $\text{ReLU}(x) = \max(0, x)$ is the rectified linear unit applied elementwise.

Fourth, we consider Model Predictive Control, called *MPC*, a model-based strategy based on (1) that optimizes control actions over a receding horizon using an approximate model of both the BESS dynamics and the IS market. We first explore a deterministic setting wherein point forecasts are used to approximate future prices. Then, a scenario-based MPC approach considers a finite set of representative realizations to maximize the expected profit.

Finally, we introduce a deterministic optimization benchmark, called *Perfect foresight*. This benchmark computes the optimal BESS schedule under the unrealistic assumption of complete knowledge of future imbalance prices. It therefore serves as a theoretical upper bound on achievable profits, as no practical or causal control strategy can outperform a scheduler with full access to future price information.

V. CASE STUDY

The case study is performed on actual data (2018-2020), obtained from Elia, the Belgian transmission system operator [42]. The years 2018 and 2019 are used for training and validation (for RL-based agents), and the entire year 2020 is used for out-of-sample testing.

All approaches from Section IV are compared using an ex-post, out-of-sample evaluation. Specifically, at each time step within the testing horizon, the decisions p_t^* from each approach are evaluated against the values of the system imbalance (SI), taking into account the agent's actions, which determine the resulting imbalance price λ_t^{SI} . Then, we compute the true ex-post profit of the BESS across the entire test period.

The study considers a 20 MW/20 MWh BESS with a round-trip efficiency of 90%. The BESS degradation costs are detailed in Section III-A4. For the NN-based actors and critics, a hyperparameter optimization was performed to select their optimal complexity. The optimal architecture for the three RL agents is given in Table I. It should be noted that, for the safe agents, the optimization layer is added on top of these architectures. Our best architecture for the actor is our LSTM-based encoder-decoder model with 1 LSTM layer composed of 64 hidden units. The encoder processes a sequence of 12 past time steps to extract temporal dependencies. The decoder leverages a look-ahead horizon of 10 future time steps.

A. Comparison analysis

Table II summarizes the performance of each decision agent on the testing dataset. Because RL-based training is stochastic, each agent is trained independently ten times. The rows "Mean agent" and "Max agent" report the performance of, respectively, the average and best-performing models among ten independently trained agents. Each model is evaluated over the entire test set, and profits are expressed as daily averages. For the *MPC* and *Perfect foresight* benchmarks, which produce deterministic results for identical inputs, the mean and maximum values are therefore identical. Reported training times

TABLE I
RL AGENTS HYPERPARAMETERS

Hyperparameter	SAC (Enc-Dec)	TD3 (Enc-Dec)	DQL (FCNN)
Actor network	64 units	64 units	–
Critic network	64 units	64 units	(500, 500)
Discount factor γ	0.99	0.99	0.99
Target update τ	10^{-2}	10^{-3}	10^{-3}
Replay buffer size	2×10^4	5×10^3	5×10^3
Batch size	256	256	256
Actor learning rate	10^{-3}	10^{-3}	–
Critic learning rate	10^{-3}	10^{-3}	10^{-3}
Actor/critic update period	4	4	–
Alpha learning rate	10^{-3}	–	–

and proportions of infeasible actions are averaged across all test days and training runs. It is worth emphasizing that the decision time of our safe RL agent is below 2 milliseconds. Other RL-based agents have even lower computing times, while the MPC remains under 0.5 seconds. This makes it feasible to deploy these agents in RT operations.

1) *Comparison of MPC with RL-Based Agents:* As shown in Table II, the pure model-based MPC is consistently outperformed by all RL-based agents in terms of profitability. While MPC achieves a daily profit of 4,782 €/day on the test set, the average SAC and TD3 agents reach 6,012 €/day and 5,312 €/day, respectively. The safety-enhanced variants perform even better, with the average SafeSAC and SafeTD3 achieving 8,132 €/day and 8,093 €/day.

To better understand the factors behind this performance gap, Table III analyzes how the daily profit achieved by the MPC strategy varies with the look-ahead horizon (ranging from $H = 4$ to 40 time steps) and the number of forecasted price scenarios.

With a short look-ahead horizon of only 4 time steps, the MPC consistently yields negative profits, regardless of the number of input scenarios. This indicates that short-sighted decisions are insufficient, regardless of the forecast diversity. In contrast, profits increase substantially when the look-ahead is extended to 20 time steps (i.e., 5 hours), beyond which performance stabilizes, showing no significant gains when the horizon is further extended to 10 hours.

We also see that MPC is highly sensitive to the number of input scenarios. Since MPC optimizes future actions based on predicted trajectories, any error in these forecasts directly propagates to the decision quality, often resulting in suboptimal actions. Hence, relying on a single deterministic forecast of future SI leads to a lack of robustness. This translates into overoptimistic decisions that fail to account for uncertainty and often incur substantial penalties. In this way, for the optimal horizon of 40 steps, the daily profit increases from 3,774 €/day with a single scenario to 4,782 €/day with five scenarios, which is a 26.7% increase, which illustrates the benefits of incorporating forecast diversity.

In contrast, the proposed RL approach learns risk-aware trading strategies directly from historical market data, enabling it to navigate uncertainty without relying on explicit price forecasts or simplified behavioral models of other participants. By exploiting the statistical regularities present in observed

price and volume trajectories, the policy develops adaptive decision patterns that remain robust and deployable in real-world markets where the actions of competing agents are inherently unobservable.

2) *Comparison of Safe RL with Unconstrained RL Agents:* Traditional RL agents inherently struggle to infer physical dynamics and operational limits of BESS during training. As a result, unconstrained agents tend to explore infeasible or suboptimal regions of the action space, leading to degraded performance and policy instability. In practice, these agents frequently propose infeasible actions, violating SOC or power limits in up to 21 % and 27 % of cases for SAC and TD3, respectively.

By failing to guarantee the safety of the actions, unconstrained RL agents tend to provide overoptimistic decisions (that may not be applicable in practice due to BESS constraints). Consequently, all safety-enhanced RL agents achieve higher profitability and feasibility. For instance, the mean test profit of the unconstrained SAC agent reaches 6,012 €/day, while its safety-informed variants, SAC_{pen}, SAC_{rew}, and SafeSAC, improve performance to 7,246 €/day, 7,368 €/day and 8,132 €/day, respectively.

3) *Comparison of Projection-Based Safe RL with Soft-Constrained RL Agents:* Among the safety-informed agents, SafeSAC and SafeTD3 exhibit the best outcomes. In this way, from all benchmarks, SafeSAC reaps the highest gains with average and maximum test profits of respectively 8,132 €/day and 9,283 €/day. This is mainly due to the fact that it directly focuses on the true objective $J(\mathcal{M}_\theta)$ and is not biased by arbitrary penalization terms. Indeed, modifying the reward signals (as done by SAC_{rew}, SAC_{pen} and TD3 equivalents) helps informing the agent away from infeasible actions, but it also introduces bias into the return function, which can mislead the agent toward suboptimal regions of the policy space. If unsafe actions are heavily penalized, the agent may become overly conservative, avoiding valuable high-reward strategies. Conversely, too weak penalties may cause the agent to adopt overoptimistic policies that frequently violate feasibility constraints. Here, SAC_{pen} reaches an average test profit of 7,246 €/day, which is 11 % lower than SafeSAC. We also see that SAC_{pen} and SAC_{rew}, which rely on soft constraints, exhibit similar performances.

Overall, the projection towards the safe space ensures that SafeSAC learns in the most relevant area of the input and action spaces. As a result, the proposed SafeSAC has a better ability to limit downside risk, which can lead to better risk-adjusted performance than unconstrained or soft-constrained RL policies. Interestingly, by guiding learning toward feasible and informative trajectories, SafeSAC also accelerates convergence and stabilize policy updates. This is illustrated in Fig. 5 that compares the training speed of SafeSAC, SafeTD3, and the traditional SAC. The lines represent the average training performance over ten independent runs, while the shaded areas indicate the corresponding standard deviation, reflecting the variability in convergence behavior.

We see that SafeSAC not only converges towards higher ex-post return but also faster than the traditional SAC. The results also show that SafeSAC and SafeTD3 achieve comparable final

TABLE II

AVERAGE TESTING PERFORMANCE OF ALL DECISION AGENTS OVER TEN INDEPENDENT TRAINING RUNS. “MEAN AGENT” AND “MAX AGENT” DENOTE THE MODELS ACHIEVING, RESPECTIVELY, THE MEAN AND HIGHEST TEST PROFITS ACROSS THESE RUNS.

Metric	Model-based		Safety-informed Actor-Critic						Actor-Critic		Value-based
	<i>Perfect foresight</i>	MPC	<i>SafeSAC</i>	<i>SACpen</i>	<i>SACrew</i>	<i>SafeTD3</i>	<i>TD3pen</i>	<i>TD3rew</i>	<i>SAC</i>	<i>TD3</i>	<i>DQL</i>
Mean agent - test profit (€/day)	20,517	4,782	8,132	7,246	7,368	8,093	7,401	7,214	6,012	5,312	3,474
Max agent - test profit (€/day)	20,517	4,782	9,283	8,346	8,789	8,933	8,312	8,291	6,880	5,924	3,917
Average training time (h)	—	—	5.47	1.72	1.73	5.18	1.71	1.71	1.72	1.71	1.98
Average infeasible actions [%]	0	0	0	2.2	2	0	2	1.8	21	27	35

performance, but differ in their convergence behavior. *SafeSAC* converges after 20,000 episodes while *SafeTD3* needs around 35,000 episodes. This can be explained by the more robust exploration of *SafeSAC* arising from its stochastic policy, leading to faster and more stable learning in practice.

The best-performing *SafeSAC* agent achieves an average daily profit of approximately 9,283 €/day, corresponding to about 45% of the 20,517 €/day profit obtained by the *Perfect foresight* benchmark. This performance is promising given the intrinsic volatility and poor predictability of imbalance prices. Indeed, the *Perfect foresight* agent assumes complete knowledge of future prices, i.e., an unrealistic condition in practice, particularly in the IS where short-term price forecasts are highly uncertain. In contrast, *SafeSAC* learns directly from historical market data to develop adaptive policies that ensure feasible operation and maintain a balanced trade-off between safety and profitability under real-world conditions.

4) *Comparison Between DQL and Continuous RL Agents:* Finally, for the value-based *DQL* algorithm, we found that the highest ex-post profits are achieved when $\mathcal{A}$ is discretized

into $n = 20$ bins. Beyond this threshold, the benefits of finer granularity seem to be compensated by the challenges associated with the growing dimensionality of the Q -network output layer, which makes learning more difficult.

It can be seen in Table II that, even at its optimal discretization level, *DQL* only reaches a maximum test profit of 3,917 €/day, compared to 5,924 €/day for *TD3* and 6,880 €/day for *SAC*. Indeed, these actor-critic algorithms are inherently designed for continuous action spaces, and benefit from parameterized policies that enables smooth decisions.

B. Distributional Analysis of Profit and Risk Across Agents

To complement the previous analysis with a finer view of daily profit variability across the test set, Fig. 6 presents the distribution of daily profits for the best-performing *SafeSAC* agent (selected among the ten training runs) and three benchmark methods.

The profits are normalized by the optimal profit achieved by the *Perfect Foresight* agent. This normalization facilitates direct comparison across methods with different absolute profit scales and highlights variability even though MPC profits remain substantially lower than those of the RL-based approaches. Indeed, although the average normalized profit of *MPC* appears similar to that of *SafeSAC* and *SACpen*, it is important to recall from Table II that its absolute average profit is significantly lower, i.e., only 4,782 €/day, compared to 9,283 €/day for *SafeSAC* and 8,346 €/day for *SACpen*. This gap arises because MPC performs relatively well on low-volatility, low-profit days but suffers large losses during high-volatility periods when forecast errors are more severe.

The results show a wider dispersion in MPC outcomes, reflecting its strong dependence on forecast accuracy. In contrast, the RL-based agents learn directly from experience, enabling them to better exploit volatile and hard-to-predict imbalance prices. Fig. 6 also highlights that *SafeSAC* achieves higher mean and median profits, as well as improved performance in the upper quantiles, indicating its ability to approach near-optimal profit levels.

C. Ablation Study on Agent Architectures

Table IV presents an ablation study evaluating the added value of the key architectural and training components from the *SafeSAC* agent. We already showed in Table II that incorporating the safety layer increases the mean ex-post profit from 6,012 €/day (for the unconstrained *SAC*) to 8,132 €/day

TABLE III
PROFIT (€/DAY) FOR MPC WITH VARYING LOOK-AHEADS AND INPUT SCENARIOS.

Look-ahead (LA)	1 scenario	3 scenarios	5 scenarios
4	-1,870	-2,914	-1,938
20	3,621	4,204	4,750
40	3,774	4,257	4,782

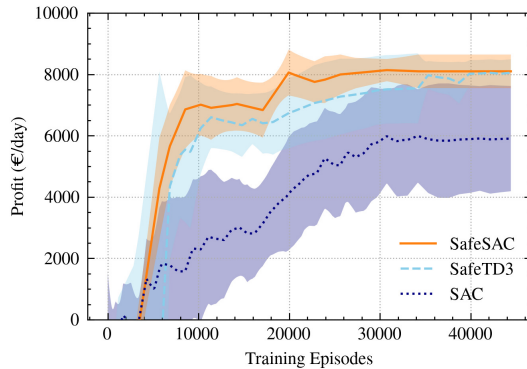


Fig. 5. Training convergence of *SafeSAC*, *SafeTD3* and *SAC* agents. The safe agents have good convergence properties, enabled by the safety layer that ensures feasibility throughout the learning process.

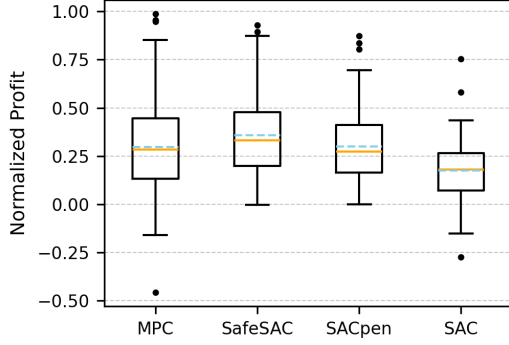


Fig. 6. Boxplots of daily profits normalized by the perfect-foresight optimum, allowing for a clearer visual comparison between methods; orange lines indicate median values, and blue lines denote means.

(for *SafeSAC*). Here, we further investigate the role of other components by testing the following variants:

- *SafeFC-NN*: a fully connected NN agent without recurrent connections across time steps, but using the same enriched input space as *SafeSAC* (including past observations and future forecasts). This configuration isolates the effect of temporal memory.
- *FC-NN*: a fully connected NN without recurrence and without the safety layer, used as a baseline to show the effect of feasibility enforcement relative to *SafeFC-NN*.
- *Markov FC-NN*: a baseline agent using SoC, calendar features, and the next-step forecast of imbalance prices, without historical inputs. Unlike *SafeFC-NN*, it follows the Markov assumption and excludes temporal context.
- *SafeSAC no EG*: the full *SafeSAC* is retained, but the expert-guided exploration introduced in Section III-D2 is removed to quantify its impact on policy learning.

TABLE IV
ABLATION ANALYSIS SHOWING THE PERFORMANCE OF THE *SafeSAC* ON THE FULL TEST SET FOR DIFFERENT NN ARCHITECTURES.

Metric	<i>SafeSAC</i>	<i>SafeFC-NN</i>	<i>FC-NN</i>	<i>Markov FC-NN</i>	<i>SafeSAC no EG</i>
Mean agent - profit (€/day)	8,132	7,994	7,753	4,797	6,408
Max agent - profit (€/day)	9,283	9,212	8,218	5,212	9,117
Training time (h)	5.47	1.21	0.32	0.28	5.46
Infeasible actions [%]	0	0	2.9	29	0

As reported in Table IV, the *Markov FC-NN* achieves an average profit of 4,797 €/day, while the best model among ten training runs reaches 5,212 €/day, outperforming the 40-step look-ahead *MPC* benchmark (4,782 €/day). Extending the state representation to include historical data and multi-step forecasted imbalance information further improves performance to 7,753 €/day for the *FC-NN* architecture, showing the importance of adding temporal context to the agent.

Incorporating the safety projection layer further improves performance to 7,994 €/day for *SafeFC-NN*, confirming that eliminating infeasible actions yields higher ex-post profits. Finally, introducing temporal recurrence through the proposed *SafeSAC* architecture raises the mean profit to 8,132 €/day. The 1.7 % improvement over *SafeFC-NN* quantifies the added

value of dealing with partial observability, showing that the recurrent *SafeSAC* agent better captures latent temporal dependencies and operational relationships.

We also observe that imitating expert actions during the early stages of training prevents the agent from being stuck in a local optimum. Without expert guidance, the agent may prematurely converge to a suboptimal policy that exploits immediate rewards by discharging energy early and be stuck with an empty BESS. In contrast, expert-provided trajectories expose the agent to more profitable long-term strategies, such as accepting negative immediate rewards when purchasing energy in order to achieve higher future returns. Expert trajectories provide an initial approximation of the long-term policy, which the agent subsequently refines through its own experience.

D. Sensitivity Analysis on Forecast Accuracy

Forecast accuracy is a critical factor for any BESS control strategy, as errors directly propagate into suboptimal or infeasible decisions. To assess this effect, we conduct a dedicated sensitivity analysis on the impact of forecast errors.

We generate perturbed versions of the original forecasted SI price signals, $\lambda_t^{SI,q}$ (as defined in Subsection III-A1). The perturbations are created by adding controlled levels of Gaussian noise [43]. These noisy signals, which are used as inputs to the *SafeSAC* agent, are defined as:

$$\hat{\lambda}_t^{SI,q} = \lambda_t^{SI,q} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2) \quad (25)$$

where $\lambda_t^{SI,q}$ is the original forecasted SI price for quantile q , $\hat{\lambda}_t^{SI,q}$ is the perturbed forecast value, and σ_ϵ is the standard deviation controlling the amplitude of the noise ϵ . Three forecast error scenarios are considered: (i) low-error forecasts ($\sigma_\epsilon = 10\%$ of the median price), (ii) medium-error forecasts ($\sigma_\epsilon = 50\%$), and (iii) high-error forecasts ($\sigma_\epsilon = 100\%$), covering both realistic and extreme conditions.

Table V reports the mean daily profit, and the standard deviation σ_{p^*} of the selected actions p^* over the test set, which serves as an indicator of policy conservativeness. A lower σ_{p^*} indicates a more cautious policy that limits energy exchanges between the BESS and the grid. Both metrics are averaged over 10 random realizations of the noise ϵ_t for the *SafeSAC*, *SACpen*, *SAC*, and *MPC* agents.

Under low forecast noise ($\sigma_\epsilon = 10\%$), the profit of the *SafeSAC* agent decreases by only 0.3%, compared to a 5% drop for *MPC*. With moderate noise ($\sigma_\epsilon = 50\%$), *SafeSAC*

TABLE V
MEAN DAILY PROFITS AND STANDARD DEVIATIONS σ_{p^*} OF ACTIONS p_t^* OVER THE TEST SET, FOR DIFFERENT FORECAST NOISES, AVERAGED OVER 10 DIFFERENT PERTURBED REALIZATIONS PER AGENT.

σ_ϵ (%)		<i>SafeSAC</i>	<i>SACrew</i>	<i>SAC</i>	<i>MPC</i>
10	Profit (€/day)	8,109	7,271	5,891	4,558
	σ_{p^*} (MW)	11.4	12.2	13.4	16.2
50	Profit (€/day)	6,598	6,432	4,912	3,344
	σ_{p^*} (MW)	10.2	10.2	10.6	16
100	Profit (€/day)	5,008	5,029	4,117	1,863
	σ_{p^*} (MW)	7.6	7.2	8.4	16.2

retains 81% of its original profit, while *MPC* falls to 70%. At high noise levels ($\sigma_\epsilon = 100\%$), *SafeSAC* still preserves 62% of its initial profit, whereas *MPC* declines sharply to just 39%.

This analysis highlights that forecast accuracy affects optimization-based methods like *MPC* more strongly, as prediction errors directly influence their decisions. In contrast, RL-based agents are less sensitive, as return-guided training enables them to learn policies that are more robust to imperfect information. This adaptive conservativeness is reflected in how the distribution of agents' action p_t^* change with varying levels of forecast accuracy. As forecast noise σ_ϵ increases, these agents reduce the amount of energy exchanged with the grid, as a risk-mitigation strategy. For instance, the variability σ_{p^*} of the actions of *SafeSAC* drops from 11.4 MW at 10% noise to 7.6 MW at 100%. In contrast, the standard deviation σ_{p^*} of *MPC* actions remains nearly constant across all noise levels (around 16.2 MW), suggesting that its decisions do not adapt to the reliability of forecasts.

E. Learning Feasibility Constraints via the Projection Layer

The lightweight design of the proposed safety layer enables its seamless use during both training and deployment, ensuring that final actions p_t^* are always feasible. To assess how much the policy depends on this layer, and how well it learns to internalize BESS constraints, we evaluate the raw (pre-projection) actions $\hat{P}_t$ produced by the agent. Fig. 7 reports the evolution of infeasible action rates (based on $\hat{P}_t$) and test-set profits for *SafeSAC*, alongside *SACrew* and *SAC*, averaged over 10 training runs.

Both *SafeSAC* and *SACrew* agents exhibit a strong decline in infeasible actions early in training. For *SafeSAC*, this aligns with rising profits, indicating that learning feasibility directly enhances economic efficiency. Then, *SafeSAC* continues to improve even after the feasibility stabilizes, indicating more effective exploration within the feasible region. This highlights the dual role of the projection layer, i.e., not only enforcing feasibility but also guiding the agent toward higher-quality decisions and faster convergence. In contrast, the standard *SAC* agent demonstrates slower and less consistent learning of BESS constraints, as it lacks any explicit mechanism for feasibility and relies solely on naive action clipping. Similarly, *SACrew* reduces infeasible actions early on through penalty terms in the reward, but without truly internalizing the constraints. As a result, both agents yield lower ex-post profits compared to *SafeSAC*.

Interestingly, Fig. 7 shows that the raw (pre-projection) actions $\hat{P}_t$ of the trained *SafeSAC* agent are largely feasible, with only about 4 % violating BESS constraints. This suggests that the agent has learned to internalize most of the feasibility structure within its NN policy during training, rather than depending entirely on the projection layer. The projection thus acts as a lightweight safeguard that enforces strict constraint satisfaction while allowing the policy to concentrate on optimizing long-term returns. Yet, these results highlight that keeping the projection layer at deployment remains essential: it ensures full feasibility of p_t^* without introducing overhead or sacrificing performance.

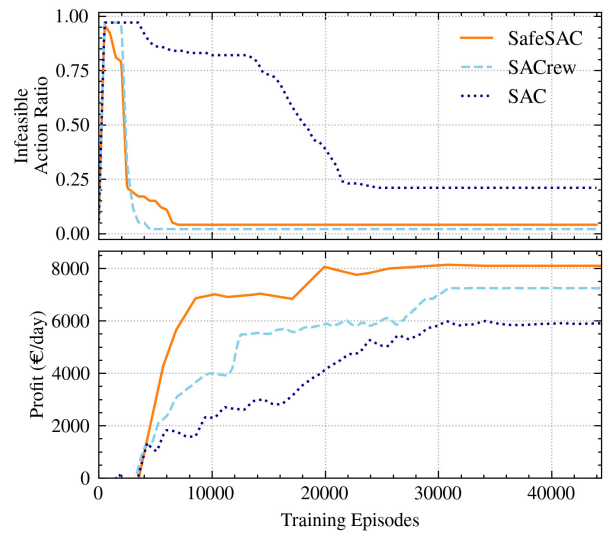


Fig. 7. Evolution of test profit and infeasible action ratio among *SafeSAC* (considering the raw action $\hat{P}_t$), *SACrew* and *SAC* agents, averaged over 10 training runs. The *SafeSAC* agent not only relies on the projection layer but internalizes the physics comparably to the *SACrew*.

VI. CONCLUSIONS

This paper addresses the operational strategy of a BESS participating in the real-time IS market. The proposed methodology combines the strengths of both RL, capable of effectively addressing the high level of stochasticity in the IS market, with model-based optimization to enforce known techno-economic constraints. This is achieved through a safe RL agent where recurrent NN layers capture market dynamics under partial observability, while a differentiable projection layer enforces physical BESS feasibility, enabling end-to-end training and robust (data-consistent) profit maximization in the IS market.

On our case study using Belgian data on a specific BESS, this hybrid approach exhibits an ex-post profit superior of 10% compared to other safe-augmented RL methods. Our safe RL also achieves a 70% and 35% performance improvement over the traditional model predictive control, and the corresponding pure RL agent, respectively.

Analyses show that enriching the feature space with past information and future forecasts is essential for informed decisions, while a recurrent architecture further improves handling of partial observability in the IS market. In addition, benchmark schedules such as model predictive control provide valuable training trajectories for our off-policy algorithm, enhancing learning stability.

Future work will focus on exploring rolling online RL updates and post-training knowledge distillation to efficiently adapt to market shifts with minimal retraining cost. Another promising direction is to extend the framework to multi-asset settings, where coordinated operation under joint feasibility constraints can be optimized.

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